

Research Project Summary

EndSurvGAN: End-to-End Survival-Aware Generative Modeling of Censored Time-to-Event Data

Zahra Esfandiar

University of Tehran

z.esfandiar214@gmail.com

Overview

This project introduces **EndSurvGAN**, an end-to-end survival-aware generative framework for synthetic censored time-to-event data generation. The main objective of the project is to generate realistic synthetic survival datasets while preserving the joint relationships among covariates, survival times, and censoring indicators.

Existing synthetic tabular data generators often treat survival time and censoring indicators as ordinary variables. As a result, they may fail to preserve key survival-specific structures such as censoring mechanisms, survival distributions, and the dependency between covariates and event times. In contrast, EndSurvGAN was designed to directly learn the joint survival distribution and generate synthetic samples that preserve both statistical structure and survival dynamics.

Research Motivation

Access to high-quality survival datasets is often restricted by privacy concerns, ethical limitations, and data-sharing regulations, especially in healthcare and biomedical research. Synthetic data generation provides a useful direction for privacy-preserving data sharing, reproducible research, model benchmarking, and simulation-based evaluation of survival methods.

However, censored survival data are more complex than ordinary tabular data because each observation contains not only covariates, but also an observed time and an event indicator. Therefore, a useful synthetic survival generator should preserve:

- the marginal and joint distribution of covariates,
- the relationship between covariates and survival times,
- censoring patterns,
- survival curves and time-to-event structure,
- downstream predictive utility for survival models.

EndSurvGAN was developed to address these challenges through a unified survival-aware generative modeling framework.

Problem Formulation

Let

$$D = \{(x_i, t_i, \delta_i)\}_{i=1}^N$$

denote a survival dataset, where $x_i \in \mathbb{R}^d$ is the covariate vector, $t_i \in \mathbb{R}^+$ is the observed survival time, and $\delta_i \in \{0, 1\}$ is the event indicator.

Under right censoring, the observed time and event indicator are defined as

$$T = \min(Y, C),$$

and

$$\Delta = \mathbb{I}(Y \leq C),$$

where Y denotes the latent event time and C denotes the censoring time.

The objective of EndSurvGAN is to learn the joint distribution

$$P(X, T, \Delta)$$

and generate synthetic samples

$$(x^*, t^*, \delta^*) \sim P_G(X, T, \Delta),$$

where P_G denotes the distribution learned by the generator. The generated samples should preserve the statistical structure, survival dynamics, and censoring behavior of the original data.

Model Architecture

The EndSurvGAN architecture consists of a generator network and a discriminator network trained under an adversarial learning framework.

The generator receives a latent noise vector z and produces a synthetic survival sample:

$$G(z) = (x^*, t^*, \delta^*),$$

where x^* represents generated covariates, t^* represents generated survival time, and δ^* represents the generated censoring indicator.

The discriminator receives either a real survival sample

$$s = (x, t, \delta)$$

or a generated survival sample

$$s^* = (x^*, t^*, \delta^*)$$

and learns to distinguish real samples from synthetic samples.

Unlike standard GAN-based tabular data generators, EndSurvGAN incorporates survival-specific regularization terms to ensure that generated data preserve both survival outcomes and censoring patterns.

Training Objective

The total objective function combines adversarial learning with survival-aware and censoring-aware regularization:

$$\mathcal{L}_{total} = \mathcal{L}_{adv} + \lambda_{surv} \mathcal{L}_{surv} + \lambda_{cens} \mathcal{L}_{cens},$$

where:

- $\mathcal{L}_{adv}$ is the adversarial loss.
- $\mathcal{L}_{surv}$ is the survival-consistency regularization term.
- $\mathcal{L}_{cens}$ is the censoring-aware regularization term.
- λ_{surv} and λ_{cens} control the contribution of each regularization component.

For stable adversarial training, the model adopts the Wasserstein GAN with Gradient Penalty objective:

$$\mathcal{L}_{adv} = \mathbb{E}_{s^* \sim P_G}[D(s^*)] - \mathbb{E}_{s \sim P_{data}}[D(s)] + \lambda_{gp} \mathbb{E}_{\hat{s}} (\|\nabla_{\hat{s}} D(\hat{s})\|_2 - 1)^2,$$

where $\hat{s}$ denotes interpolated samples between real and generated survival observations, and λ_{gp} is the gradient penalty coefficient.

To preserve the relationship between covariates and survival outcomes, the survival-consistency loss is defined as

$$\mathcal{L}_{surv} = D_{KL}(P_{data}(T|X) \parallel P_G(T^*|X^*)).$$

Similarly, the censoring-aware loss is defined as

$$\mathcal{L}_{cens} = D_{KL}(P_{data}(\Delta|X) \parallel P_G(\Delta^*|X^*)).$$

These regularization terms encourage the generator to reproduce both realistic survival times and realistic censoring mechanisms.

Technical Implementation

I implemented the complete EndSurvGAN framework in **Python** using **PyTorch**. The implementation included the full model architecture, custom loss functions, training loop, hyperparameter configuration, evaluation pipeline, and ablation study.

The technical implementation included:

- Construction of the generator and discriminator networks.
- Design of the survival-aware regularization module.
- Design of the censoring-aware regularization module.
- Implementation of the WGAN-GP optimization framework.
- Survival data preprocessing and encoding.
- Training and validation pipeline.
- Kaplan–Meier curve comparison.
- Distributional fidelity evaluation.
- Downstream predictive utility assessment using the C-index.
- Ablation analysis to evaluate the contribution of each model component.

The generator and discriminator were implemented as fully connected neural networks with three hidden layers and 256 units per layer. Training was performed using the Adam optimizer with learning rate 2×10^{-4} , batch size 256, latent dimension 128, and WGAN-GP regularization coefficient $\lambda_{gp} = 10$. All models were trained for 200 epochs.

Experimental Evaluation

EndSurvGAN was evaluated on three benchmark survival datasets: **SUPPORT**, **ACTG**, and **Rotterdam**. These datasets differ in sample size, number of features, and censoring rate, allowing evaluation across different survival-data settings.

Dataset	Samples	Features	Censoring Rate
SUPPORT	9,105	14	32%
ACTG	2,467	17	64%
Rotterdam	2,982	7	57%

The proposed method was compared against Gaussian Copula, TVAE, CTGAN, and SurvivalGAN. Evaluation metrics included distributional fidelity, Kaplan–Meier distance, and downstream predictive utility measured using the Concordance Index.

Key Results

Across all benchmark datasets, EndSurvGAN achieved improved performance compared with competing tabular and survival-specific generative models.

Dataset	Method	KS/JS	KM-Distance	C-index
SUPPORT	SurvivalGAN	0.101	0.086	0.68
SUPPORT	EndSurvGAN	0.082	0.061	0.71
ACTG	SurvivalGAN	0.095	0.081	0.69
ACTG	EndSurvGAN	0.078	0.058	0.72
Rotterdam	SurvivalGAN	0.097	0.079	0.71
Rotterdam	EndSurvGAN	0.081	0.063	0.74

The results indicate that EndSurvGAN better preserves the joint survival structure, improves Kaplan–Meier curve alignment, and provides stronger downstream predictive utility.

Ablation Study

An ablation study was conducted to evaluate the contribution of the proposed survival-aware and censoring-aware regularization components.

Model Variant	C-index
Two-Stage Generator	0.67
EndSurvGAN without Censoring Loss	0.69
EndSurvGAN without Survival Loss	0.68
Full EndSurvGAN	0.71

The full model achieved the strongest performance, suggesting that both survival-aware and censoring-aware regularization contribute to improved synthetic survival data generation.

Research Skills Developed

This project strengthened my experience in:

- Survival analysis and censored data modeling.
- Generative adversarial networks.
- Statistical learning and deep learning.
- Computational statistics and optimization.
- PyTorch model development.
- Custom loss function design.
- Survival-specific evaluation metrics.
- Methodological research for biomedical data.

Future Directions

Future extensions of EndSurvGAN include competing-risk survival data, recurrent-event generation, privacy-preserving synthetic data generation, and uncertainty-aware survival modeling.

These directions align with my broader research interests in statistical methodology, high-dimensional data analysis, predictive modeling, and computational statistics.

Publication Status

Submitted to the 18th Iranian Statistics Conference (2026). Review process ongoing.